

VIT Bhopal University

School of Computing Science and Engineering

COMPUTER VISION PROJECT REPORT

HelioScan

Drone-Based Photovoltaic Panel Defect Detection & IEC 62446-3 Degradation Severity Mapping

A Modular, Zero-GUI Headless Computer Vision Engineering System

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1. Introduction

Utility-scale photovoltaic (PV) plants contain large numbers of modules that remain exposed to thermal cycling, ultraviolet radiation, mechanical impact, soiling and other environmental stresses throughout their operating life. These conditions can lead to hotspots, microcracks, Potential-Induced Degradation (PID), soiling and bypass-diode failures.

HelioScan addresses this inspection problem through an automated, headless Computer Vision pipeline designed for aerial RGB/thermographic imagery. It processes images, isolates and localizes photovoltaic defects, computes deterministic severity scores, derives a Module Degradation Index (MDI), assigns operational health tiers, and exports structured telemetry.

2. Problem Statement

Existing inspection workflows can face three practical limitations:

- **GUI dependency:** interactive desktop environments are unsuitable for autonomous drone companion computers and headless cloud execution.
- **Subjective scoring:** simple visual or color thresholds do not directly incorporate physical defect area, thermal differential or electrical proximity.
- **False positives:** PV cell grid lines and metallic busbars can resemble defect patterns and reduce detection reliability.

HelioScan therefore provides a reproducible zero-GUI pipeline with explicit defect classes, physical severity formulation and structured engineering telemetry.

3. Objectives

- Automate photovoltaic module inspection from UAV imagery.
- Detect five defect categories: Hotspots, Microcracks, PID, Soiling and Bypass Diode Failure.
- Use image-processing operations including LAB-CLAHE, bilateral filtering, morphology, thresholding and Non-Maximum Suppression.
- Compute a bounded defect severity score S_i in $[0, 100]$.
- Aggregate defect severity into a Module Degradation Index (MDI: 0-100).
- Provide health tiers and maintenance-oriented recommendations.
- Run completely headlessly and export JSON/CSV engineering telemetry.

4. Functional Requirements

ID	Requirement
FR-1	Image ingestion, validation, LAB-CLAHE, bilateral filtering and perspective rectification.
FR-2	Multi-class detection of Hotspots, Microcracks, PID, Soiling and Diode Failures.
FR-3	Deterministic severity calculation using area, temperature differential and busbar proximity.
FR-4	MDI calculation and Health Tier 1-4 classification.
FR-5	Headless HUD rendering with JSON and CSV telemetry export.

5. Non-Functional Requirements

- **Performance:** target processing above 15 FPS with less than 65 ms per frame.
- **Headless operability:** terminal execution without X11, Wayland or graphical window managers.
- **Reliability:** validation and exception handling for malformed images, coordinates and inputs.
- **Maintainability and testability:** modular source structure and automated pytest validation.

6. System Architecture

Stage 1 - Ingestion & Preprocessing: image validation -> LAB CLAHE -> bilateral filtering -> module ROI/perspective extraction.

Stage 2 - Defect Detection: thermal anomaly thresholding -> directional morphology -> contour classification -> NMS.

Stage 3 - Severity Engine: relative defect area + temperature differential + busbar proximity + hazard weight.

Stage 4 - Health Analyzer: MDI calculation -> Health Tier assignment -> maintenance recommendation.

Stage 5 - Visualization & Export: OpenCV HUD overlay -> JSON telemetry -> CSV engineering log.

The implementation is modularized into preprocessing, detection, severity, analysis, metrics, visualization and pipeline components, supported by test and utility files in the GitHub repository.

7. Design Diagrams & Data Contracts

The VITyarthi guidelines require architecture/workflow documentation and applicable UML diagrams, together with dataset description and evaluation methodology for computation-heavy or ML-related projects.

Field	Type / Meaning
image_stem	String - unique inspection image identifier
defect_id	Integer - per-image defect index
category	Enum - Hotspot, Microcrack, Soiling, Diode_Failure, PID
confidence	Float - detection certainty, 0.0-1.0
bbox_[x,y,w,h]	Integer - bounding-box pixel coordinates
temp_delta_c	Float - calibrated thermal differential
severity_score	Float - 0.0-100.0
module_mdi	Float - 0.0-100.0
health_tier	Enum - Tier 1 to Tier 4

8. Design Decisions & Rationale

- **CIELAB CLAHE:** applying contrast enhancement to the perceptual lightness channel avoids directly equalizing RGB chromatic channels.
- **Bilateral filtering:** preserves important spatial edges while reducing high-frequency noise, supporting narrow defect structures.
- **1D column smoothing:** suppresses narrow busbar-like spikes while retaining broader heating patterns.
- **Zero-GUI architecture:** allows terminal-based deployment on drone companion computers and headless servers.

9. Implementation Details & Mathematical Formulation

Defect Severity:

$$S_i = \min(100.0, w_c \times [\alpha(A_i/A_{\text{cell}})100 + \beta(\Delta T_i/\Delta T_{\text{max}})100 + \gamma P_{\text{busbar}}])$$

where $\alpha = 0.35$, $\beta = 0.45$, $\gamma = 0.20$ and $\Delta T_{\text{max}} = 50$ deg C. Hazard weights are: Diode Failure 1.50, PID 1.40, Hotspot 1.30, Microcrack 1.15 and Soiling 0.85.

Module Degradation Index:

$$\text{MDI} = \max(0.0, 100.0 - [0.60 \times \max(S_i) + 0.40 \times \text{mean}(S_i)] \times [1 + 0.12(N - 1)])$$

Estimated power loss is defined as $P_{\text{loss}} = (100 - \text{MDI}) \times 0.68\%$.

Operational Health Tiers

Tier	MDI	Action
1 - Optimal	≥ 85	Continue routine UAV flights.
2 - Degraded	70 to <85	Schedule washing/re-inspection within 30 days.
3 - Compromised	50 to <70	Dispatch field technician to test bypass diodes.
4 - Critical	<50	Immediate string isolation and panel replacement.

10. Experimental Benchmark & Results

Scenario	GT	Pred.	Precision	Recall	F1	Latency
clean_module_01.jpg	0	0	1.000	1.000	1.000	225.4 ms
hotspot_01.jpg	1	1	1.000	1.000	1.000	25.3 ms
hotspot_02.jpg	2	2	1.000	1.000	1.000	25.9 ms
microcrack_01.jpg	1	1	1.000	1.000	1.000	32.7 ms
soiling_01.jpg	1	1	1.000	1.000	1.000	28.7 ms
diode_failure_01.jpg	1	1	1.000	1.000	1.000	26.8 ms

Overall Precision	100.00%
Overall Recall	100.00%
Overall F1-Score	100.00%
Mean IoU	0.759
Average Processing Latency	60.8 ms/frame

These figures are the benchmark results supplied for the project and represent the included representative sample set rather than a claim of generalization to all real-world PV imagery.

11. Testing Approach & Quality Assurance

- Preprocessing tests validate loading, dimensions, CLAHE, bilateral filtering and ROI extraction.
- Detector tests cover clean-module false positives, hotspot/soiling detection and NMS deduplication.
- Severity tests validate score bounds, weighting and tier behavior.
- Analyzer tests validate clean MDI, degradation penalties and critical alarms.
- Metrics tests cover IoU, Precision, Recall and F1.
- Pipeline tests validate end-to-end single-image and batch execution.

Reported test result: 23 passed in 0.82 s (100% pass rate).

12. Challenges Faced & Solutions

- **Busbar and grid-line false positives:** directional length filters suppress continuous straight structures crossing cell boundaries.
- **Aluminum-frame glare:** a 16-pixel internal margin is used before thermal column averaging.
- **Variable ambient irradiance:** adaptive CLAHE in perceptual CIELAB space improves contrast handling.

13. Learnings & Key Takeaways

- Applied OpenCV morphology, CIELAB/HSV color transformations and Non-Maximum Suppression.

- Translated an engineering inspection requirement into deterministic Computer Vision rules and mathematical scoring.
- Understood the practical role of headless execution for autonomous and resource-constrained inspection systems.

14. Future Enhancements

- Quantized ONNX/TensorRT models for real-time NVIDIA Jetson Orin deployment.
- GPS/georeferenced 3D digital-twin or GIS integration for asset-level anomaly mapping.
- Night-time electroluminescence imaging support for improved microscopic microcrack inspection.

15. References

1. International Electrotechnical Commission, IEC TS 62446-3: Photovoltaic systems - Part 3: Outdoor infrared thermography of modules and plants, 2017.
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4. ASTM International, ASTM D6433: Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys, 2020.